AGENT EVALUATION
SCIENCE

Evaluating Agent Swarms: Token Budgets, Active Workers, and Baseline Choice

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Abstract

Evaluating an agent swarm requires distinguishing the effects of increasing the total token budget, dividing a fixed ceiling, and changing coordination. We examine these comparisons in 8,400 controlled trials on fresh arithmetic and graph-colouring tasks. At 4,096 completion tokens per worker, increasing independent width from one to four improves accuracy by 16.0 percentage points on both families while the total ceiling quadruples. Splitting a fixed 4,096-token completion ceiling instead lowers graph accuracy from 56.0% to 18.5%; the arithmetic change is inconclusive. The primary comparisons do not establish an accuracy advantage over a sequential control matched on scheduled calls and nominal token ceiling. Ring communication provides no consistent gain. Eleven complementary scaling studies add 20,376 held-out trials across seven task families, including a fresh-task sixteen-worker confirmation that does not establish superiority over fixed-four execution. Candidate and request audits show that finding a valid answer after an initial failure depends on the token allowance. They also reveal incomplete worker activation and 97 controlled-study attempts with unknown usage. The findings show how resource controls, output selection, and execution records qualify claims about agent scaling. They support evaluating specific allocation policies; equal-realized-compute advantages and general scaling laws remain unestablished.

Keywords: Agent evaluation; Multi-agent systems; Test-time compute; Token budgets; Resource allocation; Coordination; Reliability; Reproducibility

> "All happy families are alike; each unhappy family is unhappy in its own way." —Leo Tolstoy, *Anna Karenina* (Tolstoy, 2014).
>
> *Adapted to this study:* All successful swarms share a valid outcome; unsuccessful swarms fail in different ways.

1 Introduction

Adding agents changes more than the number of contributors. Under a shared token ceiling, wider systems receive smaller per-worker allowances, generate more candidates, and may use different amounts of the available budget. A nominal worker can also fail to become active. An observed accuracy difference therefore measures the deployed allocation and selection procedure, not an isolated effect of collaboration.

This paper asks what fixed-ceiling worker-count evaluations can establish about task success, resource efficiency, and communication. We examine eleven completed studies on programming, logical reasoning, graph colouring, arithmetic search, and planning. Three measurement distinctions organize the analysis: a ceiling versus realized token use; requested versus active workers; and improvement over one worker versus improvement over a strong fixed-width baseline. These distinctions matter even when task grading is deterministic.

Our contribution is an empirical audit of these distinctions in saved configuration grids, including negative results, feasibility diagnostics, and a fresh-task policy confirmation. We report all 176 held-out cells, retain repeated tasks as statistical clusters, and make the limits of the to-

ken ledger explicit. The resulting reporting recommendations are derived from the audit; they have not themselves been validated as an evaluation standard.

Relation to prior work. Repeated sampling separates solution coverage from successful candidate selection (Brown et al., 2024); self-consistency and debate provide different aggregation and interaction mechanisms (Wang et al., 2023; Du et al., 2023). Effective inference allocation depends on task difficulty (Snell et al., 2024), and controlled agent-system studies already show dependence on architecture and task structure (Kim et al., 2026). We do not claim task dependence as a new principle. Shared-workspace communication can help discovery tasks given sufficient resources and feedback (Park et al., 2026); our single-round ring comparison tests a narrower intervention. Failure taxonomies (Cemri et al., 2025) complement our resource and activation diagnostics, which do not by themselves identify a causal failure mechanism.

2 Evaluation Design

Figure 1 shows the controlled schedules and token allowances. The complementary studies below retain their original allocations and selection protocols.

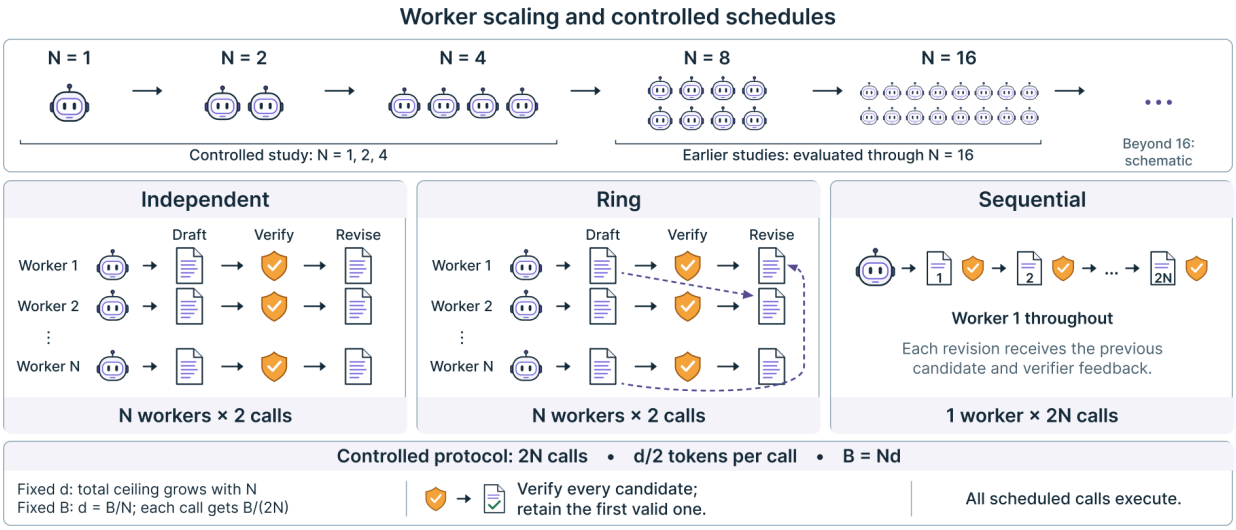


Figure 1 | Worker scaling and controlled schedules. N denotes parallel width, d the per-worker completion allowance, and $B = Nd$ the total completion ceiling. All controlled schedules execute $2N$ calls, each allowed $d/2$ completion tokens. The controlled study tests $N = 1, 2, 4$; earlier studies extend through $N = 16$. Widths beyond sixteen are schematic.

Evidence and task coverage. Table 1 summarizes the eleven complementary studies. Public task sources are BigCodeBench (Zhuo et al., 2025), MBPP with EvalPlus tests (Austin et al., 2021; Liu et al., 2023), ZebraLogic (Lin et al., 2025), BBEH (Kazemi et al., 2025), and PlanBench (Valmeekam et al., 2023). Graph-colouring and Countdown instances are generated algorithmically and scored by deterministic constraint checks; graph generation is inspired by Reasoning Gym (Stojanovski et al., 2025). These are fixed subsets or adaptations, not full-benchmark leaderboard scores. Luna 5.6, Luna 6, and Muse Contributor are recorded model identifiers; they are not interchangeable model versions (Appendix A).

Interventions and selection. The original programming and ZebraLogic grids evaluate $N \in \{1, 2, 4, 8, 16\}$ at three ceilings. The five smaller diagnostic studies compare one and four workers at two ceilings. The confirmation evaluates a transferred policy and its controls. Workers

Task family / study	Models	Studies	Tasks*	Eval. trials
BigCodeBench	L5.6, M	2	32	3,456
MBPP+	L6, M	2	16	864
ZebraLogic original	L6	1	64	3,456
ZebraLogic confirmation	L6	1	300	6,600
Custom graph colouring	L6, M	2	100	2,400
Custom Countdown	L6	1	100	1,200
BBEH subset	L6	1	100	1,200
PlanBench subset	L6	1	100	1,200
Total		11		20,376

Table 1 | Completed evaluation evidence. L5.6/L6: Luna 5.6/6; M: Muse Spark Contributor. *Distinct evaluation tasks per study; task IDs can overlap between model studies. MBPP+ uses one repetition per task and configuration; all others use two. An additional 6,272 calibration trials give 26,648 total. Pilot, smoke, imported baseline, incomplete, and superseded records are excluded.

draft and revise; ring revision adds a predecessor's draft to the revision context. Independent revision omits that peer draft. Both conditions can still share a final integrator in the programming and ZebraLogic workflows. Thus the contrast does not remove every form of information sharing.

Graph colouring, Countdown, and PlanBench select a locally valid revision when available. Their width effects include verifier-assisted parallel search. BBEH instead uses gold-blind plurality with deterministic tie breaking. Programming and ZebraLogic use a model integrator, with final grading outside the generation loop. Appendix A records allocations and graders.

Outcomes and resource accounting. For a complementary study, nominal ceiling B, width N, and revision condition c, let $\widehat{S}_{N,c}(B)$ denote mean final-task success and $\overline{T}_{N,c}(B)$ mean recorded tokens. We compare each wider cell to its shared $N = 1$ control:

$$\Delta S = 100\big(\widehat{S}_{N,c}(B) - \widehat{S}_1(B)\big), \qquad R_T = \frac{\overline{T}_{N,c}(B)}{\overline{T}_1(B)}. \tag{1}$$

ΔS is in percentage points (pp). T retains each study's charged-token or retained-debit convention, including conservative debits where applicable. It is not a complete ledger of all attempted calls or provider compute. Equal ceilings do not imply equal realized resources. Original calibration also contains disclosed accounting overruns; nominal ceilings are not a claim of universally verified hard caps (Appendix B). These charged-token and retained-debit ceilings include prompt accounting; Figure 1 instead specifies a completion-token ceiling for the controlled schedules.

Uncertainty and analysis units. Cells share tasks, repetitions, and controls. Paired bootstrap intervals resample whole tasks, keeping their observations together and using the study-specific strata and draw counts in Appendix C. The 26,648 trials are not independent task samples. Configuration counts describe the evaluated grids; they do not estimate a population frequency or constitute 148 independent hypothesis tests. Reported 95% intervals are study-specific and are not adjusted for cross-study multiplicity.

3 Findings and Their Evaluation Consequences

3.1 A shared ceiling does not equalize resource use

All 148 wider held-out cells use more recorded tokens than their matched-ceiling control. Accuracy is lower in 75, tied in 24, and higher in 49 (Figure 2A). Thus 99 cells have higher recorded token use without a higher mean success rate. Programming contributes 96 wider cells, 81

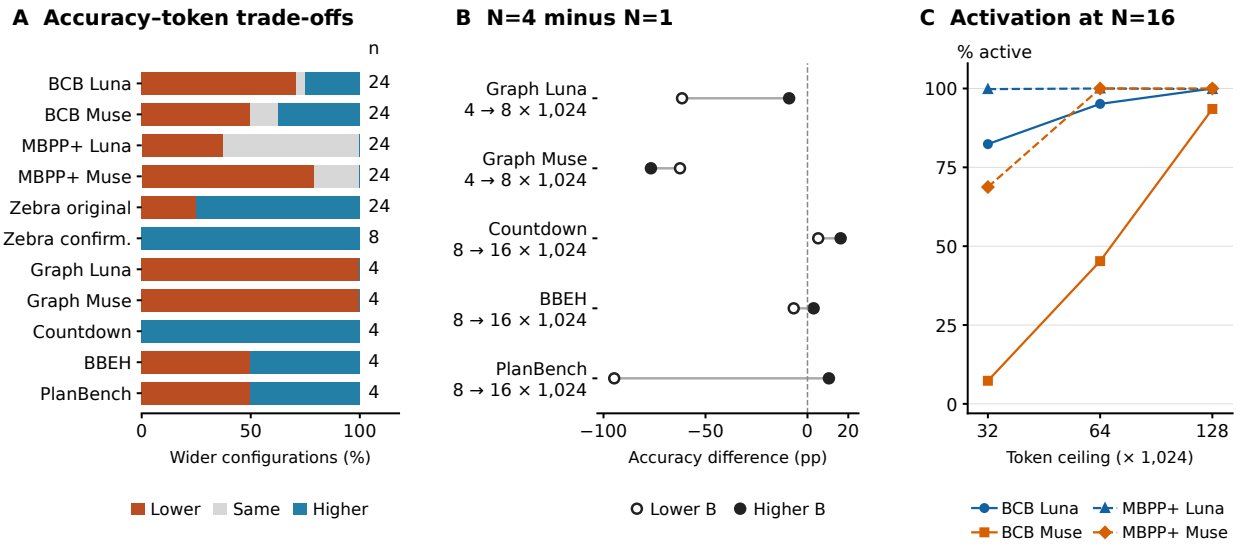


Figure 2 | Three views of the saved evaluation evidence. (A) Fractions of wider cells with lower, equal, or higher accuracy than matched-ceiling $N = 1$; every wider cell uses more recorded tokens ($n = 148$ configurations, not independent samples). (B) Four-worker accuracy effects, pooled over revision conditions, in the five two-ceiling diagnostic studies. (C) Recorded activation, pooled over revision conditions, in the four programming studies that report it. Complete cell values and paired communication intervals appear in Appendices C–D.

of them flat or worse; the overall count reflects unequal grid sizes and must not be read as a benchmark-balanced effect.

The finding is a joint statement about observed accuracy and the recorded resource ledger. It does not establish population-level dominance for every cell, nor does it compare policies at equal realized expenditure. For example, independent four-worker Countdown gains 9.5–17.0 pp, but uses 2.77–3.85 times the control's recorded tokens. Reporting only accuracy at the shared ceiling would omit that resource difference.

3.2 Width can cross a feasibility boundary

On the PlanBench subset, independent four-worker accuracy is 0% at $B = 8{,}192$, versus 95% for one worker. At $B = 16{,}384$, the same widths score 100% and 89.5%. Their paired differences are -95.0 pp ($[-98.0, -91.5]$) and $+10.5$ pp ($[6.0, 15.5]$). Four-worker length-limit finishes fall from 99.94% to 0.19% of retained stage calls. The higher-ceiling gain uses 4.06 times the control's retained debit.

BBEH provides a complementary case without verifier-assisted selection: ring four-worker accuracy changes from a 6.5-pp loss ($[-11.5, -2.0]$) to a 4.5-pp gain ($[0.5, 9.0]$) when its ceiling doubles. Conversely, graph-colouring four-worker cells lose 7.5–77.5 pp across both models and ceilings. The appropriate interpretation is conditional feasibility under the tested allocation. Width and per-worker allowance change together; lower truncation accompanies the reversals but is not isolated as their cause.

Requested width also differs from recorded participation (Figure 2C). Pooling revision conditions, BigCodeBench Muse's sixteen requested workers average 7.3% activation at the smallest ceiling, rising to 45.3% and 93.5% at the next two ceilings. A worker-count label therefore does not identify a stable amount of realized participation. This diagnostic does not establish that activation alone explains accuracy.

3.3 Sampling gains do not establish communication gains

Countdown improves with independent workers while ring revision reduces average accuracy by 5.0 pp ($[-8.5, -1.75]$). Luna graph colouring also has a negative ring contrast, -2.0

pp ($[-4.0, -0.25]$). Most other communication intervals include or touch zero (Appendix C). PlanBench's saved stratified interval is degenerate because its only differing task occupies a singleton stratum; an unstratified sensitivity interval spans $[0.0, 0.75]$ pp. It is not evidence of general communication superiority.

These contrasts concern the tested ring intervention. Verifier-assisted candidate selection, model integration, and peer exchange are distinct mechanisms. A wider system's gain cannot be assigned to collaboration merely because multiple workers contributed candidates.

3.4 Fresh tasks require a strong baseline as well

The original, post-start-amended ZebraLogic study reports a selected-policy gain of 3.78 pp ($[1.30, 6.51]$) against its calibration-selected fixed-width baseline, which selected one worker. The confirmation transfers the communicating $4/8/16$ policy without refitting to 300 unused puzzles and compares it with communicating fixed four. Averaged over the two higher ceilings, adaptive minus fixed-four accuracy is $+0.92$ pp ($[-0.58, 2.50]$).

The original and confirmation contrasts have different comparators and cannot be pooled. The confirmation establishes neither superiority nor equivalence. Fresh tasks address task reuse, while a competitive fixed-width control addresses whether adaptation adds value. Both checks are necessary before interpreting a selected allocation rule as an established advantage.

4 Reporting Implications

The audit motivates reporting the resource and decision process alongside the final score. Table 2 connects each recommendation to an observed ambiguity. These are practical requirements for interpreting this class of comparison, not an empirically validated new metric.

Observed ambiguity	Report with the score	Comparison it enables
Same ceiling, different use	Nominal cap, realized ledger, retries, unknown usage, cap violations	Separate cap-constrained performance from efficiency at matched resources.
Width changes execution	Requested/active workers, per-stage allowances, skips and truncation	Identify participation and feasibility changes.
Multiple sources of gain	Peer context, integrator, selector and grader access	Separate independent search from the tested communication change.
Policy gain depends on control	Frozen mapping, selection data, fresh tasks, single-worker and fixed-width controls	Test whether adaptation improves on a competitive deployment rule.
Repeated tasks and sparse strata	Task counts, paired resampling, stratum sizes, amendments	Avoid treating repeated runs as independent evidence.

Table 2 | Reporting specification derived from the observed evaluation problems. Additional fields diagnose ambiguity; recording them alone does not remove confounding or establish causal mechanisms.

A direct next experiment should vary width and per-worker allowance independently, keep selection and feedback access fixed, and measure the accuracy–resource frontier. Frozen adaptive policies should then face both single-worker and fixed-width controls on new tasks with matched realized resources. A larger ceiling alone is insufficient to make that comparison.

5 Limitations and Reproducibility

The studies are heterogeneous, use unequal grids, and are not replications from independent laboratories. Programming samples contain only 16–32 evaluation tasks per study. Public-task exposure and unknown training contamination limit generalization. Deterministic grading does not establish that a task subset represents deployed agent work. Most protocols are short draft–revision workflows rather than long-horizon, tool-using environments.

Model versions, provider routes, tier verification, and historical accounting differ. Usage from lost calls and retry expenditure can be unknown. The original ZebraLogic study and confirmation include disclosed operational amendments; the evidence is not uniformly preregistered. Several pilot truncation warnings were acknowledged before the frozen diagnostic grids continued. We retain those failures rather than selecting only feasible cells. No scaling exponent, exact compute comparison, latency advantage, or causal effect of truncation is established.

The appendices document protocols, amendments, interval definitions, and every evaluation cell for the eleven complementary studies. Their normalized saved outcomes, concise protocol notes, provenance hashes, and aggregate verification code are available in a versioned public repository.[1] These materials support reproduction of the reported grid summaries, not complete inference replay or independent validation of every grader. No new model runs were performed for this reframing.

6 Conclusion

Worker-count comparisons are interpretable only with their allocation, participation, selection, and baseline conditions. In these eleven studies, extra workers often consume more recorded tokens without improving mean accuracy, budget changes can reverse width effects, and a fresh-task policy confirmation leaves its advantage over fixed four unresolved. Reporting these conditions turns an apparent worker-scaling result into a more specific, testable claim about an evaluated agent system.

[1] https://github.com/PluvioXO/agent-swarms-reproducibility/tree/reproducibility-v1.1

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A Study Inventory and Protocols

This appendix documents the eleven complementary studies represented in the normalized evidence export. It omits prospective scaling models, unfinished experiments, and imported full-benchmark baselines. Study codes below identify the complete cell table in Appendix D and the machine-readable source bundle. Counts are completed scientific trial keys, not request attempts.

Code	Study	Cal. tasks	Cal. trials	Eval. tasks	Eval. trials
BC-L	BigCodeBench Luna 5.6	32	1,792	32	1,728
BC-M	BigCodeBench Muse	32	1,792	32	1,728
MP-L	MBPP+ Luna 6	16	448	16	432
MP-M	MBPP+ Muse	16	448	16	432
Z-O	ZebraLogic original	32	1,792	64	3,456
Z-C	ZebraLogic confirmation	0	0	300	6,600
GC-L	Graph colouring Luna 6	0	0	100	1,200
GC-M	Graph colouring Muse	0	0	100	1,200
CD	Countdown Luna 6	0	0	100	1,200
BB	BBEH subset Luna 6	0	0	100	1,200
PB	PlanBench subset Luna 6	0	0	100	1,200
	Total		6,272		20,376

Table A1 | Scientific trial inventory. Every non-MBPP+ cell has two repetitions per task; MBPP+ has one. Calibration and evaluation tasks are distinct within the original studies. The two graph-colouring studies reuse the same tasks; cross-model task counts must not be added as independent samples. Separate pilots and smoke checks are not calibration trials in this inventory.

A.1 Grids, models, and information flow

The five original full-grid studies (BC-L, BC-M, MP-L, MP-M, Z-O) use calibration ceilings of 8,192, 16,384, 32,768, and 65,536 tokens, with widths 1, 2, 4, and 8. The single-worker control is shared between conditions, giving seven cells per ceiling and 28 calibration cells per study. Evaluation uses ceilings 32,768, 65,536, and 131,072 with widths 1, 2, 4, 8, and 16, giving nine cells per ceiling and 27 evaluation cells. Draft, revision, and model integration receive nominal budget shares of 50%, 25%, and 25%; draft and revision shares are divided across workers. Minimum output reserves and prompt costs can prevent calls from launching.

BC-L records `gpt-5.6-luna`; later Luna studies request `openai/gpt-6-luna` through OpenRouter, generally with low reasoning and requested Flex service. Muse studies record `meta/muse-spark-1.3-contributor`. Original BigCodeBench sampling records differ between Luna and Muse; the latter's stored Flex label is not proof that Flex was requested or served. The graph-colouring Muse runner does not request a service tier and uses provider-default reasoning. These experiments do not form a single pinned-model, pinned-provider study.

The graph-colouring studies use ceilings 4,096 and 8,192; Countdown, BBEH, and PlanBench use 8,192 and 16,384. At each ceiling, they compare one independent worker, four independent workers, and four workers with ring revision. Draft and revision each receive 50% of the total allowance, divided equally over workers. There is no model-integration stage in these five studies. Ring revision receives a predecessor draft; independent revision receives the worker's own draft. Recorded activation is available only for BC-L, BC-M, MP-L, and MP-M. We use their stored `active_N` field rather than inferring participation from requested width or API concurrency.

A.2 Task construction and confirmation

Graph colouring uses 100 locally generated greedy-solvable three-colour graphs, stratified over 10, 12, 14, 16, and 18 vertices, with edge probability 0.2 and generator seed 20260928.

Studies	Final candidate construction	Final task criterion
BigCodeBench	Model integration of worker revisions	Saved sandbox test outcome; no hidden-test feedback to workers.
MBPP+	Model integration of worker revisions	Base and EvalPlus tests.
ZebraLogic	Model integration of worker revisions	Normalized whole-grid exact correctness; malformed/empty grids fail.
Graph colouring	First locally valid revision in worker order; otherwise first nonempty revision; no draft rescue	Local three-colour validity for the generated graph.
Countdown	Verifier-assisted selection of a valid candidate	Local arithmetic constraints and target match.
BBEH subset	Gold-blind plurality of normalized answers; worker-order tie breaking	Local normalized exact/fuzzy matching to the reference answer.
PlanBench subset	First locally valid revision in worker order	Local STRIPS plan validity; optimality is not required.

Table A2 | Selection and grading are distinct parts of the evaluated system. Deterministic local grading is not independent validation of grader correctness or benchmark representativeness.

The generator is inspired by Reasoning Gym but is not its official task distribution. Countdown has 100 algorithmically generated custom puzzles with four to eight supplied numbers. Both generated task families use deterministic correctness checks; valid colourings or arithmetic expressions need not be unique. BBEH selects 25 tasks from each of disambiguation QA, multistep arithmetic, spatial reasoning, and time arithmetic from `google-deepmind/bbeh`, revision `80d12ca916b7158f22293fcf3144f4d3d854d4be`. PlanBench selects 100 Blocksworld Hard instances from `harshakokel/PlanBench`, revision `1f4d600f4806bedbefebbbe44b168372aaf5060d`, and uses a local interpreter. BBEH and PlanBench ring revision receives a predecessor excerpt capped at 512 estimated tokens. Their independent revisions receive the worker's own draft. No full-benchmark or unaided-collaboration score is inferred from these subsets.

The original ZebraLogic data are pinned to `WildEval/ZebraLogic`, `grid_mode`, revision `0a473f5a0054835754ed156d5a79c6ce27178bb1`. Its independent power policy selects widths 2/2/2 across the evaluation ceilings; its communicating power policy selects 4/8/16. The calibration-selected fixed-width baseline is one worker in both conditions. Its primary contrast averages the selected power policy minus that fixed-width baseline across ceilings and conditions.

The confirmation transfers the communicating 4/8/16 mapping to 300 unused puzzles, with 12 puzzles per recorded size stratum, without refitting. At the smallest ceiling it evaluates one worker and independent/ring four; at the middle ceiling it adds independent/ring eight; at the highest it adds independent/ring sixteen. Shared cells are deduplicated, yielding 11 cells and 6,600 trials. The primary contrast averages ring eight minus ring four at 65,536 and ring sixteen minus ring four at 131,072. It is not an adaptive-versus-single-worker contrast.

B Resource Accounting, Failures, and Provenance

The retained token variable is copied from each completed trial's `tokens_accounting.total_charged_tokens` or `charged_tokens`. Depending on the study, these fields combine reported usage, local estimates, and conservative full-allocation debits for eligible unknown-usage calls. They are neither an exact physical-compute measure nor a complete account of operational expenditure. Ratios in the paper compare cells within the same study and ceiling; we do not pool token totals across models to estimate universal compute efficiency.

Twelve included BC-M calibration trials exceed their nominal ceilings, by at most 1.92%, under the historical accounting implementation. No included evaluation trial's recorded total

exceeds its nominal ceiling in the normalized data. This check covers the retained ledger; it cannot establish that unobserved attempts or provider-side work respected those ceilings.

The diagnostic studies retain stage finish reasons. PlanBench's independent four-worker cells contain 1,600 retained stage calls at each ceiling: 1,599 finish at the length limit at 8,192, compared with three at 16,384. The corresponding single-worker cells contain 400 calls each and no length-limit finishes. These observations support a feasibility diagnostic; they do not identify truncation as the sole causal explanation for the accuracy reversal. Skips, refusals, and incomplete outputs can have other causes.

Operational amendments and excluded records. The original ZebraLogic protocol was amended after work began. Its record discloses revised gates and scheduling, conservative usage recovery, bounded provider and connection retries, and seven abandoned whole-trial attempts. The confirmation retained an additional upstream-503 retry amendment after 1,496 evaluation trials. These studies must not be described as uniformly preregistered. Lost-call usage and cost remain unknown where they were not reported; retained outcomes do not imply zero operational failure expenditure.

Graph colouring has separate 30-trial pilots per model; Countdown and PlanBench have 60 pilot trials each, and BBEH has 48. Several pilot truncation warnings were acknowledged before evaluation continued on the frozen grid. Original ZebraLogic contributes 50 separate pilot and nine smoke trials; confirmation contributes 33 smoke trials. BigCodeBench and MBPP+ smoke records are also excluded. The combined export includes imported full-benchmark baselines that are not part of this analysis. One MBPP+ Muse evaluation row marked `analysis_included=false` is superseded and excluded. We use the 316 retained calibration/evaluation cell summaries, of which 176 are evaluation cells, rather than counting every exported row.

Analysis artifacts and availability. Reproducibility materials for the eleven complementary studies are available in the versioned repository snapshot at https://github.com/PluvioXO/agent-swarms-reproducibility/tree/reproducibility-v1.1. At the repository root, `evidence/trials.csv` is a normalized export of all 26,648 retained trials with task IDs, repetitions, configurations, binary final outcomes, and recorded token debits. It also includes activation where recorded and retained stage-call/length-finish counts for the seven later studies whose phase records are present. Empty fields mean unavailable, not zero. `evidence/cell-summary.json` preserves the original saved cell summaries and communication/policy interval metadata. `verify_results.py` checks unique trial keys, all 316 cell aggregates, the descriptive counts, activation rates, and the main PlanBench diagnostics.

`evidence/provenance.json` records the SHA-256 hashes and original workspace filenames of the supplied combined experiment export, PlanBench export, manuscript sections, figure data, and reused vector figure. `docs/METHODS.md` summarizes the grids, task sources, selection rules, and paired inference; `docs/DATA.md` defines the trial fields and study counts. The repository also includes the current paper PDF and `SHA256SUMS` for file integrity. Full source/run archives, historical reports, and raw provider payloads are outside this concise peer-review package. The normalized trial data and saved cell summaries are unchanged.

The snapshot supports analysis of saved outcomes but does not claim complete inference replay, complete attempt accounting, or independent grader validation. The separate controlled study's trial records are not included in this snapshot.

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C Paired Inference and Sensitivity Analyses

Each task is a cluster containing its configurations and repetitions. Bootstrap resampling retains the full cluster so that a wider cell and its control remain paired. The programming communication contrasts use 20,000 task bootstrap draws with seed 20260921. The original ZebraLogic primary interval uses its saved 4,000 paired-task draws in manifest task order with seed 20260926; the confirmation uses 10,000 size-stratified draws with seed 20260927.

The smaller diagnostic studies use 10,000 paired task draws for their primary communication intervals. Their strata are graph size, Countdown size, BBEH category, or PlanBench object count. The per-budget BBEH and PlanBench width contrasts use 20,000 deterministic paired draws within their respective strata, retaining both repetitions for each sampled task. These are conditional, study-specific intervals; no pooled meta-analysis or cross-study multiplicity correction is applied.

Study	Ring minus independent (pp)	95% interval
BigCodeBench Luna 5.6	$+0.39$	$[-1.04, 2.34]$
BigCodeBench Muse	-1.43	$[-3.65, 0.39]$
MBPP+ Luna 6	-1.56	$[-4.17, 0.00]$
MBPP+ Muse	$+2.60$	$[0.00, 7.81]$
Graph colouring Luna 6	-2.00	$[-4.00, -0.25]$
Graph colouring Muse	-0.75	$[-2.25, 0.75]$
Countdown	-5.00	$[-8.50, -1.75]$
BBEH subset	$+1.75$	$[-0.50, 4.00]$
PlanBench subset[†]	$+0.25$	$[0.25, 0.25]$

Table C1 | Every saved within-study ring contrast, averaging the evaluated multi-worker widths, ceilings, and repetitions. ZebraLogic has distinct selected-policy contrasts, described below. [†]The PlanBench interval reflects a singleton stratum, not population-level certainty.

Sparse-stratum sensitivity. PlanBench's ring-versus-independent difference comes from one passing repetition on `planbench-hard-34`, the only evaluation task in its seven-object stratum. The stratified bootstrap retains that task in every draw, yielding a degenerate $[0.25, 0.25]$-pp interval. A post-hoc unstratified paired-task bootstrap with 50,000 draws, seed 20260929, gives $[0.00, 0.75]$ pp. Neither supports a practically meaningful general communication advantage. A degenerate stratified interval is a property of this resampling design, not evidence of zero population uncertainty.

Policy comparison and comparator sensitivity. The original ZebraLogic primary estimate is $+3.776$ pp with interval $[1.302, 6.510]$, averaging both revision conditions and all three ceilings against the calibration-selected fixed-width policy. The confirmation's primary estimate is $+0.917$ pp, interval $[-0.583, 2.500]$, against communicating fixed four at the two higher ceilings. The latter point estimate can be read directly from the saved counts: at 65,536, ring eight passes 538/600 while ring four passes 531/600; at 131,072, ring sixteen passes 542/600 while ring four passes 538/600. Averaging these two paired accuracy differences gives 11/1,200, or 0.917 pp. The intervals concern different estimands and must not be combined into a common policy effect. An interval spanning zero does not establish equivalence.

D Complete Held-Out Configuration Cells

The following table lists every evaluation cell, including all 28 shared single-worker controls and all 148 wider cells. Codes are defined in Appendix A. I means independent revision and R

means ring revision. B is the nominal total ceiling; T is mean recorded tokens (charged tokens or retained debit); ΔS and R_T use Equation (1). Pass counts are over repeated observations of the same study tasks. Values are rounded only for display.

Study	B	N	Mode	Passes	T	ΔS (pp)	R_T
BC-L	32,768	1	I	39/64	2,672	0.00	1.00
BC-L	32,768	2	I	38/64	4,828	-1.56	1.81
BC-L	32,768	2	R	41/64	5,113	$+3.12$	1.91
BC-L	32,768	4	I	41/64	8,611	$+3.12$	3.22
BC-L	32,768	4	R	39/64	9,058	$+0.00$	3.39
BC-L	32,768	8	I	40/64	14,295	$+1.56$	5.35
BC-L	32,768	8	R	40/64	14,473	$+1.56$	5.42
BC-L	32,768	16	I	44/64	15,342	$+7.81$	5.74
BC-L	32,768	16	R	42/64	15,340	$+4.69$	5.74
BC-L	65,536	1	I	41/64	2,656	0.00	1.00
BC-L	65,536	2	I	38/64	4,678	-4.69	1.76
BC-L	65,536	2	R	39/64	5,091	-3.12	1.92
BC-L	65,536	4	I	39/64	8,939	-3.12	3.37
BC-L	65,536	4	R	40/64	9,379	-1.56	3.53
BC-L	65,536	8	I	39/64	16,691	-3.12	6.28
BC-L	65,536	8	R	40/64	17,316	-1.56	6.52
BC-L	65,536	16	I	40/64	27,847	-1.56	10.48
BC-L	65,536	16	R	39/64	28,632	-3.12	10.78
BC-L	131,072	1	I	42/64	2,649	0.00	1.00
BC-L	131,072	2	I	40/64	4,726	-3.12	1.78
BC-L	131,072	2	R	40/64	4,994	-3.12	1.89
BC-L	131,072	4	I	37/64	8,858	-7.81	3.34
BC-L	131,072	4	R	39/64	9,387	-4.69	3.54
BC-L	131,072	8	I	39/64	16,986	-4.69	6.41
BC-L	131,072	8	R	38/64	17,979	-6.25	6.79
BC-L	131,072	16	I	39/64	32,550	-4.69	12.29
BC-L	131,072	16	R	40/64	33,874	-3.12	12.79
BC-M	32,768	1	I	40/64	6,827	0.00	1.00
BC-M	32,768	2	I	40/64	12,743	$+0.00$	1.87
BC-M	32,768	2	R	36/64	13,746	-6.25	2.01
BC-M	32,768	4	I	39/64	20,344	-1.56	2.98
BC-M	32,768	4	R	39/64	20,342	-1.56	2.98
BC-M	32,768	8	I	36/64	20,017	-6.25	2.93
BC-M	32,768	8	R	34/64	20,143	-9.38	2.95
BC-M	32,768	16	I	14/64	17,123	-40.62	2.51
BC-M	32,768	16	R	13/64	17,207	-42.19	2.52
BC-M	65,536	1	I	36/64	7,191	0.00	1.00
BC-M	65,536	2	I	41/64	13,189	$+7.81$	1.83

Table D1 | Continued

Study	B	N	Mode	Passes	T	ΔS (pp)	R_T
BC-M	65,536	2	R	40/64	13,565	$+6.25$	1.89
BC-M	65,536	4	I	41/64	23,826	$+7.81$	3.31
BC-M	65,536	4	R	39/64	25,108	$+4.69$	3.49
BC-M	65,536	8	I	40/64	38,355	$+6.25$	5.33
BC-M	65,536	8	R	41/64	38,700	$+7.81$	5.38
BC-M	65,536	16	I	35/64	39,676	-1.56	5.52
BC-M	65,536	16	R	33/64	39,312	-4.69	5.47
BC-M	131,072	1	I	41/64	6,912	0.00	1.00
BC-M	131,072	2	I	41/64	12,911	$+0.00$	1.87
BC-M	131,072	2	R	39/64	13,862	-3.12	2.01
BC-M	131,072	4	I	42/64	23,948	$+1.56$	3.46
BC-M	131,072	4	R	42/64	26,325	$+1.56$	3.81
BC-M	131,072	8	I	42/64	45,551	$+1.56$	6.59
BC-M	131,072	8	R	41/64	48,503	$+0.00$	7.02
BC-M	131,072	16	I	37/64	75,378	-6.25	10.91
BC-M	131,072	16	R	40/64	75,464	-1.56	10.92
MP-L	32,768	1	I	14/16	807	0.00	1.00
MP-L	32,768	2	I	13/16	1,397	-6.25	1.73
MP-L	32,768	2	R	12/16	1,480	-12.50	1.83
MP-L	32,768	4	I	13/16	2,587	-6.25	3.21
MP-L	32,768	4	R	13/16	2,735	-6.25	3.39
MP-L	32,768	8	I	13/16	4,890	-6.25	6.06
MP-L	32,768	8	R	13/16	4,870	-6.25	6.03
MP-L	32,768	16	I	13/16	4,751	-6.25	5.89
MP-L	32,768	16	R	12/16	4,743	-12.50	5.88
MP-L	65,536	1	I	13/16	807	0.00	1.00
MP-L	65,536	2	I	13/16	1,444	$+0.00$	1.79
MP-L	65,536	2	R	12/16	1,484	-6.25	1.84
MP-L	65,536	4	I	13/16	2,578	$+0.00$	3.19
MP-L	65,536	4	R	13/16	2,769	$+0.00$	3.43
MP-L	65,536	8	I	13/16	4,959	$+0.00$	6.14
MP-L	65,536	8	R	13/16	5,355	$+0.00$	6.63
MP-L	65,536	16	I	13/16	9,550	$+0.00$	11.83
MP-L	65,536	16	R	13/16	9,271	$+0.00$	11.48
MP-L	131,072	1	I	13/16	794	0.00	1.00
MP-L	131,072	2	I	13/16	1,371	$+0.00$	1.73
MP-L	131,072	2	R	13/16	1,521	$+0.00$	1.92
MP-L	131,072	4	I	13/16	2,587	$+0.00$	3.26
MP-L	131,072	4	R	13/16	2,846	$+0.00$	3.58
MP-L	131,072	8	I	13/16	4,858	$+0.00$	6.12

Table D1 | Continued

Study	B	N	Mode	Passes	T	ΔS (pp)	R_T
MP-L	131,072	8	R	13/16	5,419	$+0.00$	6.82
MP-L	131,072	16	I	13/16	9,658	$+0.00$	12.16
MP-L	131,072	16	R	13/16	10,394	$+0.00$	13.09
MP-M	32,768	1	I	13/16	1,770	0.00	1.00
MP-M	32,768	2	I	12/16	3,214	-6.25	1.82
MP-M	32,768	2	R	12/16	3,319	-6.25	1.87
MP-M	32,768	4	I	12/16	6,062	-6.25	3.42
MP-M	32,768	4	R	12/16	6,344	-6.25	3.58
MP-M	32,768	8	I	12/16	10,271	-6.25	5.80
MP-M	32,768	8	R	12/16	9,818	-6.25	5.55
MP-M	32,768	16	I	12/16	9,091	-6.25	5.13
MP-M	32,768	16	R	13/16	8,952	$+0.00$	5.06
MP-M	65,536	1	I	13/16	1,660	0.00	1.00
MP-M	65,536	2	I	12/16	3,240	-6.25	1.95
MP-M	65,536	2	R	12/16	3,556	-6.25	2.14
MP-M	65,536	4	I	13/16	5,814	$+0.00$	3.50
MP-M	65,536	4	R	12/16	6,383	-6.25	3.84
MP-M	65,536	8	I	12/16	11,231	-6.25	6.76
MP-M	65,536	8	R	13/16	11,979	$+0.00$	7.22
MP-M	65,536	16	I	12/16	19,783	-6.25	11.92
MP-M	65,536	16	R	12/16	19,338	-6.25	11.65
MP-M	131,072	1	I	13/16	1,799	0.00	1.00
MP-M	131,072	2	I	11/16	3,304	-12.50	1.84
MP-M	131,072	2	R	13/16	3,386	$+0.00$	1.88
MP-M	131,072	4	I	11/16	5,947	-12.50	3.31
MP-M	131,072	4	R	12/16	6,601	-6.25	3.67
MP-M	131,072	8	I	12/16	11,123	-6.25	6.18
MP-M	131,072	8	R	12/16	12,680	-6.25	7.05
MP-M	131,072	16	I	12/16	22,236	-6.25	12.36
MP-M	131,072	16	R	13/16	23,600	$+0.00$	13.12
Z-O	32,768	1	I	110/128	3,298	0.00	1.00
Z-O	32,768	2	I	111/128	5,818	$+0.78$	1.76
Z-O	32,768	2	R	111/128	5,823	$+0.78$	1.77
Z-O	32,768	4	I	113/128	10,055	$+2.34$	3.05
Z-O	32,768	4	R	114/128	10,414	$+3.12$	3.16
Z-O	32,768	8	I	109/128	12,800	-0.78	3.88
Z-O	32,768	8	R	108/128	11,358	-1.56	3.44
Z-O	32,768	16	I	102/128	9,832	-6.25	2.98
Z-O	32,768	16	R	98/128	9,912	-9.38	3.00
Z-O	65,536	1	I	111/128	3,304	0.00	1.00

Table D1 | Continued

Study	B	N	Mode	Passes	T	ΔS (pp)	R_T
Z-O	65,536	2	I	115/128	5,710	$+3.12$	1.73
Z-O	65,536	2	R	115/128	5,983	$+3.12$	1.81
Z-O	65,536	4	I	114/128	10,485	$+2.34$	3.17
Z-O	65,536	4	R	114/128	11,005	$+2.34$	3.33
Z-O	65,536	8	I	113/128	19,320	$+1.56$	5.85
Z-O	65,536	8	R	114/128	19,950	$+2.34$	6.04
Z-O	65,536	16	I	106/128	24,410	-3.91	7.39
Z-O	65,536	16	R	106/128	21,670	-3.91	6.56
Z-O	131,072	1	I	107/128	3,273	0.00	1.00
Z-O	131,072	2	I	114/128	5,776	$+5.47$	1.76
Z-O	131,072	2	R	114/128	5,855	$+5.47$	1.79
Z-O	131,072	4	I	114/128	10,508	$+5.47$	3.21
Z-O	131,072	4	R	114/128	11,045	$+5.47$	3.38
Z-O	131,072	8	I	113/128	20,203	$+4.69$	6.17
Z-O	131,072	8	R	115/128	21,258	$+6.25$	6.50
Z-O	131,072	16	I	117/128	37,425	$+7.81$	11.44
Z-O	131,072	16	R	117/128	39,174	$+7.81$	11.97
Z-C	32,768	1	I	498/600	3,336	0.00	1.00
Z-C	32,768	4	I	526/600	10,153	$+4.67$	3.04
Z-C	32,768	4	R	528/600	10,528	$+5.00$	3.16
Z-C	65,536	1	I	506/600	3,332	0.00	1.00
Z-C	65,536	4	R	531/600	11,083	$+4.17$	3.33
Z-C	65,536	8	I	522/600	19,316	$+2.67$	5.80
Z-C	65,536	8	R	538/600	20,123	$+5.33$	6.04
Z-C	131,072	1	I	502/600	3,369	0.00	1.00
Z-C	131,072	4	R	538/600	11,057	$+6.00$	3.28
Z-C	131,072	16	I	536/600	37,814	$+5.67$	11.22
Z-C	131,072	16	R	542/600	39,418	$+6.67$	11.70
GC-L	4,096	1	I	198/200	1,135	0.00	1.00
GC-L	4,096	4	I	76/200	3,203	-61.00	2.82
GC-L	4,096	4	R	74/200	3,227	-62.00	2.84
GC-L	8,192	1	I	200/200	1,105	0.00	1.00
GC-L	8,192	4	I	185/200	4,561	-7.50	4.13
GC-L	8,192	4	R	179/200	4,652	-10.50	4.21
GC-M	4,096	1	I	125/200	2,916	0.00	1.00
GC-M	4,096	4	I	0/200	3,677	-62.50	1.26
GC-M	4,096	4	R	0/200	3,673	-62.50	1.26
GC-M	8,192	1	I	196/200	3,062	0.00	1.00
GC-M	8,192	4	I	44/200	7,503	-76.00	2.45
GC-M	8,192	4	R	41/200	7,490	-77.50	2.45

Table D1 | Continued

Study	B	N	Mode	Passes	T	ΔS (pp)	R_T
CD	8,192	1	I	158/200	1,811	0.00	1.00
CD	8,192	4	I	177/200	5,009	$+9.50$	2.77
CD	8,192	4	R	160/200	4,867	$+1.00$	2.69
CD	16,384	1	I	155/200	1,802	0.00	1.00
CD	16,384	4	I	189/200	6,947	$+17.00$	3.85
CD	16,384	4	R	186/200	6,273	$+15.50$	3.48
BB	8,192	1	I	105/200	1,956	0.00	1.00
BB	8,192	4	I	91/200	5,486	-7.00	2.80
BB	8,192	4	R	92/200	5,508	-6.50	2.82
BB	16,384	1	I	108/200	1,951	0.00	1.00
BB	16,384	4	I	111/200	7,364	$+1.50$	3.77
BB	16,384	4	R	117/200	7,276	$+4.50$	3.73
PB	8,192	1	I	190/200	2,552	0.00	1.00
PB	8,192	4	I	0/200	7,001	-95.00	2.74
PB	8,192	4	R	1/200	6,999	-94.50	2.74
PB	16,384	1	I	179/200	2,538	0.00	1.00
PB	16,384	4	I	200/200	10,294	$+10.50$	4.06
PB	16,384	4	R	200/200	10,721	$+10.50$	4.22

Table D1 | All 176 evaluation cells. Passes lists successful trials / total trials. Tokens are rounded mean recorded debits; I and R denote independent and ring revision. Configuration rows share tasks and controls.